

Making Academia Entrepreneurial With T&M EQUIPMENT



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Solaikutty Dhanabal, academic manager with National Instruments, recalls his time as a student when his batchmates had to frequently visit IIT Madras to use a 1GHz high-speed oscilloscope for their project as their institute lacked the one. Taking note of such requirements, test and measurement (T&M) manufacturers have ventured into the academic field to provide solutions for future designers and researchers. They aim to make test and measurement instruments familiar in all the government institutes, private as well as deemed colleges. Ultimately, it is the practical work and proper testing that add value to any research project.

Making headway with upgrades

T&M instruments are used in different disciplines of engineering and research. These also find wide applications in biotechnology.

Instruments like handheld multimeters, benchtop multimeters, AC power supplies, DC power supplies, analogue and digital oscilloscopes, LCR meters and frequency counters are widely used. However, T&M equipment for research purposes are very domain-focused, say, RF-wireless, microwave, photonics, semiconductors and the like.

While there are many instruments, industry experts suggest use of upgraded ones relevant to the industry needs. This will not only give students hands-on experience in new test machines but also make them up-to-date with industry requirements. For instance, portable equipment are believed to replace the existing benchtop products.

Many T&M providers have started imparting training at academic institutions to provide students with elementary knowledge about the industry fundamentals. And as the industry advances to Industry 4.0, institutes are also preparing the students as future designers by providing them industry training early in the

semester. This would accelerate productivity and discovery as students get practical exposure to the theory included in the course.

USB modular device replacing the conventional equipment

There are projects that students need to manage irrespective of the location. In such situations, a USB portable device comes handy as it is handheld and easier to use than conventional equipment.

The concept is to connect USB modular devices like oscilloscopes, function generators, waveform generators, power supplies and digital multimeters with one single system to download the algorithm on the system and access it anywhere. Some manufacturers provide all-in-one USB oscilloscope, multimeter and workstation.

USB-powered T&M equipment are taking the market by storm as these cost less than conventional test equipment. Those featuring USB 3.0 incorporate an increased number of logic gates (five million), which allows them to give highly accurate results and serve multiple functions.

Above all, USB modular equipment reduce power consumption to a large extent. These work more like a smart mobile that can be used continuously and left in working condition for more number of days till the testing completes.

Conventional equipment like AC power supplies start emitting a large amount of heat and may get damaged down the lane if run for up to six hours a day.

A design engineer from a reputed LED light manufacturing firm cites the incidence of an AC power supply from a reputed manufacturing source breaking down—despite it having a proper fan ventilation in-built—when it was kept on for an hour a day regularly.

Other upgrades in T&M equipment for academia include improved bitrate of oscilloscopes, whether USB modular or benchtop. The sample rate has rescaled to 10MSA/s and the resolution is now 12 bits.

USB modular
equipment (Image
courtesy: www.mcd-elektronik.com)

